Moser's Harnack Inequality*

周歧

兰州大学 基础数学

Email:zhouq2020@lzu.edu.cn

Theorem 1 (Weak Harnack Inequality) *Let B_R be a ball in $\mathbb{R}^n$ and L be given by*

$$Lu = -\sum_{i,j=1}^{n} D_j(a_{ij}(x)D_i u) + c(x)u,$$

for $a_{ij} \in L^\infty(B_R)$ satisfying

$$\lambda|\xi|^2 \le a_{ij}(x)\xi_i\xi_j \le \Lambda|\xi|^2 \quad \forall x \in B_R, \xi \in \mathbb{R}^n$$

and $c \in L^q(B_R)$,for some $q > \dfrac{n}{2}$.Assume $u \in H^1(B_R)$ satisfies $u \ge 0$ and $Lu \ge f$ weakly in B_R.If $f \in L^q(B_R)$,then,for any $0 < p < \dfrac{n}{n-2}$ and any $0 < \theta < \tau < \dfrac{1}{2\sqrt{n}}$,

$$R^{-n/p} \parallel u \parallel_{L^p(B_{\tau R})} \le C\left(\inf_{B_{\theta R}} u + R^{2-n/q} \parallel f \parallel_{L^q(B_R)}\right),$$

where C is a positive constant depending only on $n, p, q, \lambda, \Lambda, R^{2-n/q} \parallel c \parallel_{L^q(B_R)}, \theta$ and τ.

Proof

Consider $R = 1$.

We will prove

$$\parallel u \parallel_{L^p(B_\tau)} \le C\left(\inf_{B_\theta} u + \parallel f \parallel_{L^q(B_1)}\right).$$

The weak supersolution u satisfies

$$\int_{B_1} (a_{ij}D_i u D_j\varphi + cu\varphi)dx \ge \int_{B_1} f\varphi dx,$$

for any $\varphi \in H_0^1(B_1)$ with $\varphi \ge 0$ in B_1.

Set $\bar{u} = u + k > 0$,for some $k > 0$ to be determined.

We now derive an equation for $\bar{u}^{-1}$.

For any $\varphi \in H_0^1(B_1)$ with $\varphi \ge 0$ in B_1,consider $\bar{u}^{-2}\varphi$ as the test function.We have

$$\int_{B_1} \left(a_{ij}D_i u\frac{D_j\varphi}{\bar{u}^2} - 2a_{ij}D_i u D_j\bar{u}\frac{\varphi}{\bar{u}^3} + \frac{cu\varphi}{\bar{u}^2}\right)dx \ge \int_{B_1} f\frac{\varphi}{\bar{u}^2}dx.$$

*椭圆方程讨论班

Since $D_i\bar{u} = D_i u$ and $D_i(\bar{u}^{-1}) = -\bar{u}^{-2}D_i\bar{u}$.Then,we obtain

$$
\begin{aligned}
&-\int_{B_1} a_{ij}D_i(\bar{u}^{-1})D_j\varphi - 2a_{ij}D_iuD_ju\frac{\varphi}{\bar{u}^3} + \frac{cu\varphi}{\bar{u}^2}dx \geq \int_{B_1} f\frac{\varphi}{\bar{u}^2}dx \\
\Longrightarrow &\int_{B_1}\left(f\frac{\varphi}{\bar{u}^2} - \frac{cu\varphi}{\bar{u}^2}\right) + a_{ij}D_i(\bar{u}^{-1})D_j\varphi + 2a_{ij}D_iuD_ju\frac{\varphi}{\bar{u}^2} \leq 0 \\
\Longrightarrow &\int_{B_1}(a_{ij}D_i(\bar{u}^{-1})D_j\varphi + \tilde{f}\bar{u}^{-1}\varphi dx \leq 0,
\end{aligned}
$$

where

$$\tilde{f} = \frac{f}{\bar{u}} - \frac{cu}{\bar{u}}.$$

Hence,$\bar{u}^{-1}$ is a nonnegative subsolution of some homogeneous equation.
Choose $k = \| f \|_{L^q}$ if f is not identical zero.Otherwise,choose an arbitrary $k > 0$ and then let $k \longrightarrow 0^+$.In both cases,we have

$$\| \tilde{f} \|_{L^q(B_1)} \leq 1 + \| c \|_{L^q(B_1)} .$$

Theorem?? implies that,for any $\tau \in (\theta, 1)$ and any $p > 0$,

$$
\begin{aligned}
&\sup_{B_\theta} \bar{u}^{-1} \leq C \| \bar{u}^{-1} \|_{L^p(B_\tau)} \\
\Longrightarrow &\sup_{B_\theta} \bar{u}^{-p} \leq C \int_{B_\tau} \bar{u}^{-p}dx,
\end{aligned}
$$

where C is a positive constant depending only on $n, q, p, \lambda, \Lambda, \tau, \theta$.Hence

$$\inf_{B_\theta} \bar{u} \geq C\left(\int_{B_\tau} \bar{u}^{-p}dx\right)^{-1/p} = C\left(\int_{B_\tau} \bar{u}^{-p}dx \cdot \int_{B_\tau} \bar{u}^{p}dx\right)^{-1/p}\left(\int_{B_\tau} \bar{u}^{p}dx\right)^{1/p}.$$

Claim 1 There exists a $p_0 > 0$,depending only on n,such that,for any $\tau < \dfrac{1}{2\sqrt{n}}$,

$$\int_{B_\tau} \bar{u}^{-p_0}dx \cdot \int_{B_\tau} \bar{u}^{p_0}dx \leq C,$$

where C is a positive constant depending only on $n, q, \lambda, \Lambda, \tau$ and $\| c \|_{L^q(B_1)}$.
proof of claim 1.
Set

$$v = \log \bar{u}.$$

For any $\varphi \in L^\infty(B_1) \cap H_0^1(B_1)$ with $\varphi \geq 0$ in B_1,consider $\bar{u}^{-1}\varphi$ as the test function.We have

$$
\begin{aligned}
&\int_{B_1} a_{ij}D_iuD_j(\bar{u}^{-1}\varphi) + cu(\bar{u}^{-1}\varphi) \geq \int_{B_1} f\bar{u}^{-1}\varphi dx \\
\Longrightarrow &\int_{B_1} a_{ij}D_iu(-\bar{u}^{-2}D_j\bar{u}\varphi) + a_{ij}D_iu\bar{u}^{-1}D_j\varphi + cu\bar{u}^{-1}\varphi \geq \int_{B_1} f\bar{u}^{-1}\varphi.
\end{aligned}
$$

Since $D_iu = D_i\bar{u}$ and $D_iv = \bar{u}^{-1}D_i\bar{u}$,then

$$
\begin{aligned}
&-\int_{B_1} a_{ij}D_ivD_jv\varphi + \int_{B_1} a_{ij}D_ivD_j\varphi \geq \int_{B_1} f\frac{\varphi}{\bar{u}} - cu\frac{\varphi}{\bar{u}} \\
\Longrightarrow &\int_{B_1} a_{ij}D_ivD_jv\varphi \leq \int_{B_1} cu\frac{\varphi}{\bar{u}} - f\frac{\varphi}{\bar{u}} + a_{ij}D_ivD_j\varphi
\end{aligned}
$$

$$\Longrightarrow \int_{B_1} a_{ij} D_i v D_j v \varphi \leq \int_{B_1} a_{ij} D_i v D_j \varphi - \int_{B_1} \tilde{f} \varphi,$$

where $\tilde{f} = \dfrac{f}{\bar{u}} - \dfrac{cu}{\bar{u}}$.
Replace φ by φ^2,we obtain

$$\begin{aligned}
&\int_{B_1} a_{ij} D_i v D_j v \varphi^2 \leq \int_{B_1} a_{ij} D_i v D_j (\varphi^2) - \int_{B_1} \tilde{f} \varphi^2 \\
\Longrightarrow &\lambda \int_{B_1} |Dv|^2 \varphi^2 \leq 2\Lambda \int_{B_1} |Dv| \varphi |D\varphi| + \int_{B_1} |\tilde{f}| \varphi^2.
\end{aligned}$$

By the **Cauchy inequality**,we have

$$\begin{aligned}
&\lambda \int_{B_1} |Dv|^2 \varphi^2 \leq \Lambda \int_{B_1} (|D\varphi|^2 + |Dv|^2 \varphi^2|) + \int_{B_1} |\tilde{f}| \varphi^2 \\
\Longrightarrow &\int_{B_1} |Dv|^2 \varphi^2 \leq C \left(\int_{B_1} |D\varphi|^2 + \int_{B_1} |\tilde{f}| \varphi^2 \right).
\end{aligned}$$

By the **Hölder inequality** and **Sobolev inequality**,we have

$$\begin{aligned}
\int_{B_1} |\tilde{f}| \varphi^2 dx &\leq \left(\int_{B_1} |\tilde{f}|^{n/2} dx \right)^{2/n} \left(\int_{B_1} \varphi^{2n/n-2} \right)^{n-2/n} \\
&\leq C \parallel \tilde{f} \parallel_{L^{n/2}(B_1)} \int_{B_1} |D\varphi|^2 dx \\
&\leq C \parallel \tilde{f} \parallel_{L^q(B_1)} \int_{B_1} |D\varphi|^2 dx \\
&\leq C \int_{B_1} |D\varphi|^2 dx.
\end{aligned}$$

Then for any $B_{2r}(y) \subset B_1$,choose φ such that supp$\varphi \subset B_{2r}(y)$,$\varphi = 1$ in $B_r(y)$ and $|D\varphi| \leq 2r^{-1}$.We obtain

$$\int_{B_r(y)} |Dv|^2 dx \leq C r^{n-2}.$$

By the **Poincarè inequality**,we have

$$\int_{B_r(y)} |v - v_{B_r(y)}|^2 dx \leq C r^2 \int_{B_r(y)} |Dv|^2 dx \leq C r^n.$$

Since $\tau < \dfrac{1}{2\sqrt{n}}$,then

$$B_\tau \subset Q_{2\tau} \subset B_{2\sqrt{n}\tau} \subset B_1.$$

For any cube Q from $Q_{2\tau}$,take $B_r(y)$ as the smallest ball containing Q,i.e.,$Q \subset B_r(y)$,then

$$B_{2r}(y) \subset B_{2\sqrt{n}\tau} \subset B_1.$$

We obtain

$$\begin{aligned}
\int_Q |v - v_Q|^2 dx &= \int_Q v^2 - 2v \cdot v_Q + v_Q^2 dx = \int_Q v^2 dx - v_Q \int_Q 2v + v_Q^2 |Q| \\
&= \int_Q v^2 dx - 2v_Q \cdot v_Q \cdot |Q| + v_Q^2 |Q| = \int_Q v^2 dx - v_Q^2 |Q|
\end{aligned}$$

$$
\begin{aligned}
&\leq \int_Q v^2 dx - v_Q^2|Q| + (v_Q - v_{B_r(y)})^2|Q| \\
&= \int_Q v^2 dx - 2v_Q|Q|v_{B_r(y)} + v_{B_r(y)}^2|Q| \\
&= \int_Q v^2 dx - 2\int_Q v \cdot v_{B_r(y)} dx + \int_Q v_{B_r(y)}^2 dx \\
&= \int_Q |v - v_{B_r(y)}|^2 dx \leq \int_{B_r(y)} |v - v_{B_r(y)}|^2 dx \\
&\leq Cr^n \leq C|Q|.
\end{aligned}
$$

By **Hölder inequality**,we have

$$
\int_Q |v - v_Q| dx \leq \left(\int_Q |v - v_Q|^2 dx\right)^{1/2} \left(\int_Q 1^2 dx\right)^{1/2} \leq (C|Q|)^{1/2}|Q|^{1/2} \leq C|Q|.
$$

By **John-Nirenberg lemma**,we have

$$
\int_{Q_{2\tau}} e^{p_0|v - v_{Q_{2\tau}}|} dx \leq C|Q_{2\tau}| \leq C.
$$

Then,

$$
\int_{B_\tau} e^{p_0(v - v_{Q_{2\tau}})} dx \leq \int_{B_\tau} e^{p_0|v - v_{Q_{2\tau}}|} \leq \int_{Q_{2\tau}} e^{p_0|v - v_{Q_{2\tau}}|} dx \leq C
$$

and

$$
\int_{B_\tau} e^{-p_0(v - v_{Q_{2\tau}})} dx \leq \int_{B_\tau} e^{p_0|v - v_{Q_{2\tau}}|} dx \leq \int_{Q_{2\tau}} e^{p_0|v - v_{Q_{2\tau}}|} dx \leq C.
$$

Since,$v = \log \bar{u}$,then

$$
\begin{aligned}
\int_{B_\tau} \bar{u}^{-p_0} dx \cdot \int_{B_\tau} \bar{u}^{p_0} dx &= \int_{B_\tau} e^{-p_0 v} dx \cdot \int_{B_\tau} e^{p_0 v} dx \\
&= \int_{B_\tau} e^{-p_0(v - v_{Q_{2\tau}})} dx \cdot \int_{B_\tau} e^{p_0(v - v_{Q_{2\tau}})} dx \leq C.
\end{aligned}
$$

claim have been proved.

• $p = p_0$.

By **Claim 1**,we have

$$
\inf_\theta \bar{u} \geq C \left(\int_{B_\tau} \bar{u}^{p_0} dx\right)^{1/p_0}.
$$

• $p \in (0, p_0)$.

By **Hölder inequality**,we have

$$
\begin{aligned}
\left(\int_{B_\tau} \bar{u}^p dx\right)^{1/p} &\leq \left(\left(\int_{B_\tau} \bar{u}^{p \cdot p_0/p} dx\right)^{p/p_0} \cdot \left(\int_{B_\tau} 1 dx\right)^{1-p/p_0}\right)^{1/p} \\
&\leq C \left(\int_{B_\tau} \bar{u}^{p_0} dx\right)^{1/p_0}.
\end{aligned}
$$

Then,

$$
\inf_{B_\theta} \bar{u} \geq C \left(\int_{B_\tau} \bar{u}^{p_0} dx\right)^{1/p_0} \geq C \left(\int_{B_\tau} \bar{u}^p dx\right)^{1/p}.
$$

• $p \in \left(p_0, \dfrac{n}{n-2}\right)$.

Claim 2 $\forall 0 < r_1 < r_2 < 1$,$0 < p_2 < p_1 < \dfrac{n}{n-2}$.We have

$$\left(\int_{B_{r_1}} \bar{u}^{p_1} dx\right)^{1/p_1} \leq C \left(\int_{B_{r_2}} \bar{u}^{p_2} dx\right)^{1/p_2},$$

where $C \sim n, \lambda, \Lambda, q, p_1, p_2.r_1, r_2$.

proof of claim 2.

Take $\varphi = \bar{u}^{-\beta}\eta^2 \in H_0^1(B_1)$,$\eta \in C_0^1(B_0)$,$\beta \in (0,1)$.Consider φ as the test function.

We have

$$\int_{B_1} a_{ij} D_i u D_j(\bar{u}^{-\beta}\eta^2) + cu\bar{u}^{-\beta}\eta^2 dx \geq \int_{B_1} f\bar{u}^{-\beta}\eta^2.$$

Since

$$\begin{aligned}
&\int_{B_1} a_{ij} D_i u(-\beta\bar{u}^{-\beta-1}\eta^2 D_j\bar{u} + 2\eta\bar{u}^{-\beta}D_j\eta) \\
=& -\int_{B_1} \beta\eta^2 a_{ij} D_i\bar{u} D_j\bar{u}\bar{u}^{-\beta-1} + 2\int_{B_1} a_{ij} D_i\bar{u} D_j\eta\eta\bar{u}^{-\beta} \\
\leq& -\lambda\beta\int_{B_1} |D\bar{u}|^2\eta^2\bar{u}^{-\beta-1} + 2\Lambda\int_{B_1} |D\bar{u}||D\eta|\eta\bar{u}^{-\beta-1/2}\bar{u}^{1-\beta/2} \\
\leq& -\lambda\beta\int_{B_1} |D\bar{u}|^2\eta^2\bar{u}^{-\beta-1} + 2\Lambda\left(\varepsilon\int_{B_1} |D\bar{u}|^2\eta^2\bar{u}^{-\beta-1} + C(\varepsilon)\int_{B_1} |D\eta|^2\bar{u}^{1-\beta}\right),
\end{aligned}$$

then,

$$\int_{B_1} f\bar{u}^{-\beta}\eta^2 - \int_{B_1} cu\bar{u}^{-\beta}\eta^2 \leq -\lambda\beta\int_{B_1} |D\bar{u}|^2\eta^2\bar{u}^{-\beta-1} + 2\Lambda\varepsilon\int_{B_1} |D\bar{u}|^2\eta^2\bar{u}^{-\beta-1} + 2\Lambda C(\varepsilon)\int_{B_1} |D\eta|^2\bar{u}^{1-\beta}.$$

Hence,

$$\int_{B_1} |D\bar{u}|^2\eta^2\bar{u}^{-\beta-1} \leq C\left(\frac{1}{\beta^2}\int_{B_1} |D\eta|^2\bar{u}^{1-\beta} + \frac{1}{\beta}\int_{B_1}\left(\frac{|f|}{k} + |c|\right)\eta^2\bar{u}^{1-\beta}\right).$$

Set $\gamma = 1-\beta \in (0,1)$ and $w = \bar{u}^{\gamma/2}$,then

$$\int_{B_1} |Dw|^2\eta^2 dx \leq C\left(\frac{1}{(1-\gamma)^2}\int_{B_1} w^2|D\eta|^2 + \frac{1}{1-\gamma}\int_{B_1}\left(\frac{|f|}{k} + |c|\right)w^2\eta^2\right).$$

By **Hölder inequality** and **Sobolev inequality** to

$$\int_{B_1}\left(\frac{|f|}{k} + |c|\right)w^2\eta^2 dx,$$

we obtain

$$\int_{B_1} |Dw|^2\eta^2 dx \leq \frac{C}{(1-\gamma)^\alpha}\int_{B_1} w^2(|D\eta|^2 + \eta^2)dx$$

or

$$\int_{B_1} |D(w\eta)|^2 dx \leq \frac{C}{(1-\gamma)^\alpha}\int_{B_1} w^2(|D\eta|^2 + \eta^2)dx.$$

Take $\eta \in C_0^1(B_1)$,$\eta \equiv 1$ in B_r,$\eta \geq 0$,supp$\eta \subset B_R$ and

$$|D\eta| \leq \frac{2}{R-r}.$$

Then,by **Sobolev inequality**,we have

$$\begin{aligned}\left(\int_{B_r} w^{2\chi}dx\right)^{1/\chi} &\leq \left(\int_{B_R} (\eta w)^{2\chi}dx\right)^{1/\chi}\\ &\leq C\int_{B_R} |D(\eta w)|^2 dx\\ &\leq \frac{C}{(1-\gamma)^\alpha}\cdot\frac{1}{(R-r)^2}\int_{B_R} w^2 dx,\end{aligned}$$

where $\chi = \dfrac{n}{n-2}$.
Hence,

$$\left(\int_{B_r} \bar{u}^{\gamma\chi}dx\right)^{1/\gamma\chi} \leq \left(\frac{C}{(1-\gamma)^\alpha}\cdot\frac{1}{(R-r)^2}\right)^{1/\gamma}\left(\int_{B_R} \bar{u}^{\gamma}dx\right)^{1/\gamma},$$

i.e.

$$\| \bar{u} \|_{L^{\gamma\chi}(B_r)} \leq \left(\frac{C}{(1-\gamma)^\alpha}\cdot\frac{1}{(R-r)^2}\right)^{1/\gamma} \| \bar{u} \|_{L^{\gamma}(B_R)} .$$

After finitely many iteration,the claim 2 have been proved.
For any $p \in \left(0, \dfrac{n}{n-2}\right)$,we have

$$\inf_{B_\theta} \bar{u} \geq C\left(\int_{B_\tau} \bar{u}^p dx\right)^{1/p}.$$

Hence,

$$\| u \|_{L^p(B_\tau)} \leq \| \bar{u} \|_{L^p(B_\tau)} \leq C\inf_{B_\theta} \bar{u} \leq C\left(\inf_{B_\theta} u + \| f \|_{L^q(B_1)}\right).$$

□

Theorem 2 (Strong Maximum Principle) *Let Ω be a bounded domain in $\mathbb{R}^n$ and L be given by*

$$Lu = -\sum_{i,j=1}^n D_j(a_{ij}(x)D_i u) + c(x)u,$$

for $a_{ij} \in L^\infty(\Omega)$ satisfying

$$\lambda|\xi|^2 \leq a_{ij}(x)\xi_i\xi_j \leq \Lambda|\xi|^2 \quad \forall x \in \Omega, \xi \in \mathbb{R}^n$$

and $c \in L^q(\Omega)$,for some $q > \dfrac{n}{2}$.Assume $u \in H^1(\Omega)$ satisfies $u \geq 0$ and $Lu \geq f$ weakly in Ω.If,for some $B_r(x_0) \subset \Omega$,

$$\inf_{B_r(x_0)} u = 0,$$

then $u = 0$ in Ω.

Proof Assume $B_{4\sqrt{n}r}(x_0)$.By **Theorem1** with $p = 1$ to u,we have

$$r^{-n} \| u \|_{L^1(B_{2r}(x_0))} \leq C\inf_{B_r(x_0)} u.$$

Consequently $u \equiv 0$ in $B_{2r}(x_0)$.
By **covering argument**,we have $u \equiv 0$ in Ω. □

Theorem 3 (Moser's Harnack Inequality) *Let B_R be a ball in $\mathbb{R}^n$ and L be given by*

$$Lu = -\sum_{i,j=1}^n D_j(a_{ij}(x)D_i u) + c(x)u,$$

for $a_{ij} \in L^\infty(B_R)$ satisfying

$$\lambda|\xi|^2 \le a_{ij}(x)\xi_i\xi_j \le \Lambda|\xi|^2 \quad \forall x \in B_R, \xi \in \mathbb{R}^n$$

and $c \in L^q(B_R)$,for some $q > \frac{n}{2}$.Assume $u \in H^1(B_R)$ satisfies $u \ge 0$ and $Lu = f$ weakly in B_R.If $f \in L^q(B_R)$,then,

$$\sup_{B_{R/2}} u \le C\left(\inf_{B_{R/2}} u + R^{2-n/q} \parallel f \parallel_{L^q(B_R)}\right),$$

where C is a positive constant depending only on n, λ, Λ, q and $R^{2-n/q} \parallel c \parallel_{L^q(B_R)}$.

Proof By **Theorem??** and **Theorem1**,we have

$$\sup_{B_{\tau R}} u \le C\left(\inf_{B_{\tau R}} u + R^{2-n/q} \parallel f \parallel_{L^q(B_R)}\right),$$

where $\tau = \frac{1}{4\sqrt{n}}$.
By **covering argument**,we have the desired result. □

Theorem 4 (C^α Estimate) *Let B_R be a ball in $\mathbb{R}^n$ and L be given by*

$$Lu = -\sum_{i,j=1}^{n} D_j(a_{ij}(x)D_iu) + c(x)u,$$

for $a_{ij} \in L^\infty(B_R)$ satisfying

$$\lambda|\xi|^2 \le a_{ij}\xi_i\xi_j \le \Lambda|\xi|^2 \quad \forall x \in B_R, \xi \in \mathbb{R}^n$$

and $c \in L^q(B_R)$,for some $q > \frac{n}{2}$.Assume $u \in H^1(B_R)$ satisfies $u \ge 0$ and $Lu = f$ weakly in B_R.If $f \in L^q(B_R)$,then $u \in C^\alpha(B_R)$,for some $\alpha \in (0,1)$,depending only on n, q, λ and Λ.Moreover,for any $r \le \frac{R}{2}$,

$$\mathrm{osc}_{B_r} u \le C\left(\frac{r}{R}\right)^\alpha \left(R^{-n/2} \parallel u \parallel_{L^2(B_R)} + R^{2-n/q} \parallel f \parallel_{L^q(B_R)}\right),$$

where C is a positive constant depending only on n, q, λ, Λ and $R^{2-n/q} \parallel c \parallel_{L^q(B_R)}$.

Proof Set,for $r \in (0, R)$,

$$M(r) = \sup_{B_r} u, \quad \text{and} \quad m(r) = \inf_{B_r} u,$$

and

$$\omega(r) = M(r) - m(r).$$

By **Theorem??**,we have $M(r) < +\infty$ and $m(r) > -\infty$,for any $r \in (0, R)$.
Moreover,

$$|u|_{L^\infty(B_{R/2})} \le C\left(R^{-n/2} \parallel u \parallel_{L^2(B_R)} + R^{2-n/q} \parallel f \parallel_{L^q(B_R)}\right).$$

We now prove,for any $r \in \left(0, \frac{R}{2}\right)$,

$$\omega(r) \le Cr^\alpha \left(R^{-n/2} \parallel u \parallel_{L^2(B_R)} + R^{2-n/q} \parallel f \parallel_{L^q(B_R)}\right).$$

Set $\delta = 2 - \frac{n}{q}$.

Since

$$L(M(r)-u)=cM(r)-f,$$

then,by applying **Theorem3** to $M(r)-u\geq 0$ in B_r,we get

$$\sup_{B_{r/2}}(M(r)-u)\leq C\left(\inf_{B_{r/2}}(M(r)-u)+r^{\delta}(M(r)\parallel c\parallel_{L^q(B_r)}+\parallel f\parallel_{L^q(B_r)})\right).$$

i.e.,

$$M(r)-m\left(\frac{r}{2}\right)\leq C\left(\left(M(r)-M\left(\frac{r}{2}\right)\right)+r^{\delta}(M(r)\parallel c\parallel_{L^q(B_r)}+\parallel f\parallel_{L^q(B_r)})\right).$$

Similarly,by applying **Theorem3** to $u-m(r)\geq 0$ in B_r,we get

$$M\left(\frac{r}{2}\right)-m(r)\leq C\left(\left(m\left(\frac{r}{2}\right)-m(r)\right)+r^{\delta}(m(r)\parallel c\parallel_{L^q(B_r)}+\parallel f\parallel_{L^q(B_r)})\right).$$

By adding their together,we obtain

$$\omega(r)+\omega\left(\frac{r}{2}\right)\leq C\left(\left(\omega(r)-\omega\left(\frac{r}{2}\right)\right)+r^{\delta}(|u|_{L^\infty(B_r)}\parallel c\parallel_{L^q(B_r)}+\parallel f\parallel_{L^q(B_r)})\right),$$

or

$$\omega\left(\frac{r}{2}\right)\leq\gamma\omega(r)+Cr^{\delta}(|u|_{L^\infty(B_r)}\parallel c\parallel_{L^q(B_r)}+\parallel f\parallel_{L^q(B_r)}),$$

for some $\gamma=\dfrac{C-1}{C+1}<1$.

By applying **Lemma??**,we obtain,for any $r\in\left(0,\dfrac{R}{2}\right]$,

$$\omega(r)\leq C\left(\frac{r}{R}\right)^{\alpha}\left(\omega\left(\frac{R}{2}\right)+R^{2-n/q}(|u|_{L^\infty(B_{R/2})}\parallel c\parallel_{L^q(B_{R/2})}+\parallel f\parallel_{L^q(B_{R/2})})\right),$$

and hence

$$\omega(r)\leq C\left(\frac{r}{R}\right)^{\alpha}\left(|u|_{L^\infty(B_{R/2})}+R^{2-n/q}\parallel f\parallel_{L^q(B_R)}\right).$$

Since

$$|u|_{L^\infty(B_{R/2})}\leq C\left(R^{-n/2}\parallel u\parallel_{L^2(B_R)}+R^{2-n/q}\parallel f\parallel_{L^q(B_R)}\right).$$

Hence,we obtain the desired result. □

Theorem 5 (Lioubille Theorem) *Let $\Omega=\mathbb{R}^n$ be the entire space and $a_{ij}\in L^\infty(\mathbb{R}^n)$ satisfy*

$$\lambda|\xi|^2\leq a_{ij}\xi_i\xi_j\leq\Lambda|\xi|^2\quad\forall x\in\Omega,\xi\in\mathbb{R}^n.$$

Suppose $u\in H^1(\mathbb{R}^n)$ satisfies

$$\int_{\mathbb{R}^n}a_{ij}D_iuD_j\varphi dx=0,$$

for any $\varphi\in H_0^1(\mathbb{R}^n)$.If u is bounded,then u is a constant.

Proof As in the proof of **Theorem4**,we have,for some $\gamma<1$,

$$\omega(r)<\gamma\omega(2r)\quad\text{for any } r>0.$$

Hence,for any $0<r<R$,we obtain

$$\omega(r)\leq C\left(\frac{r}{R}\right)^{\alpha}\omega(R),$$

where $C > 0$ and $\alpha \in (0,1)$ are constants depending only on n, λ, Λ.Since u is bounded,then

$$\sup_{R>0} \omega(R) < \infty.$$

Next we fix an $r > 0$ and let $R \longrightarrow \infty$.We conclude

$$\omega(r) = 0.$$

This holds for any $r > 0$.Therefore,u is constant. $\square$

References

[1] Qing Han,An Introduction to Elliptic Differential Equations,Preliminary Version.